Extractive Writing AI Harness for Embodied AI Research

Professor census across QS-2027 top-100 universities (NeurIPS, ICML, ICLR, CVPR, ICCV, ECCV), a style dictionary of how these labs write, a rule-following paper writer and a 12-role scientific editor - for world models, vision-language-action models, Gaussian splatting robotics, humanoids, digital twins, AI safety and 3D/4D/6D vision.

Prepared for Manish Pandey | Research snapshot 5 November 2026 | Version 1.1 | Public edition

Website: manish-genai.github.io/paper-writing-site

Measure	Value
QS-2027 institutions	102
accepted papers indexed	94,806
List A professors (5 venues)	738
List B professors (6 venues)	581
topic-relevant List B	444
evidence records	10,641

Scope statement. This is an auditable, reproducible census built from public data, not a claim that every renowned professor in the world is listed. QS ranks institutions, not people; 'renowned' is represented only by transparent indicators (venue breadth, accepted-paper counts, presentation status, topic output). Every list entry links to evidence, and gaps are reported rather than hidden.

Public edition. This edition leaves out the per-professor tables of the full edition (top professors by topic score, topic leaders, strong output outside the QS top 100, lab style corpora and the list appendices). The professor lists are partly derived from CSRankings data, whose licence (CC BY-NC-ND 4.0) does not allow sharing modified versions. Counts and charts remain. The worked example in Section 11.3 is a made-up project.

How to use this report

Your task	Start here
Find professors / labs by topic	Sections 3-4 (method and counts); the per-professor lists are in the full edition and the local app
Understand the research frontier	Section 5 (per-topic synthesis + award papers)
Learn how top labs write	Section 6 (style dictionary) and web app 'Style Dictionary'
Follow venue rules	Section 7 and web app 'Rules & Venues'
Draft and edit your paper	Sections 8 and 11 (localhost app, Claude API or Azure)
Audit a claim or a number	Section 3 and Appendix B; audit.json exports from the app

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Headings are navigation landmarks (Word: View > Navigation Pane). Links are clickable in Word and PDF.

1 Executive summary

Decision. Treat research writing as a constrained evidence transformation, not free generation. Models may propose text and critiques; deterministic gates (hashes, exact quotes, number and citation provenance, venue rules, verbatim-overlap checks) and the author decide what enters the paper. The delivered application implements this end to end and runs locally with a Claude API key, with Claude on Microsoft Foundry (Azure billing), with Azure OpenAI, with any OpenAI-compatible server, or offline in a clearly labelled mock mode.

- **Professor census.** 738 professors at QS-2027 top-100 universities have accepted main-track papers at all of NeurIPS, ICML, ICLR, CVPR and ICCV (List A); 581 of them also at ECCV (List B). 511 (List A) and 444 (List B) are topic-relevant to the requested directions. Sources: CSRankings DBLP-derived counts (589 List A entries), OpenReview self-reported professor titles (141), and the verified 2026-10-26 dossiers (8).
- **Evidence.** 10,641 ledger records: current affiliation, QS rank, venue counts, one witness paper per venue with a primary-proceedings link (OpenReview / CVF / ECVA / PMLR / NeurIPS), match basis and author position. 683 of 738 List A entries reach evidence grade E2 (all required venues linked with strong identity evidence, or a recent self-reported professor title confirming the CSRankings affiliation).
- **Data-quality discovery.** About 12-16% of OpenReview-derived records in some years carry per-author profile fields (position, affiliation, DBLP id) in a different order from the author list. Left unchecked this produced errors such as a senior Berkeley professor appearing as an assistant professor elsewhere. The harness now detects permutations from homepage evidence, discards misaligned metadata, and learns DBLP ids only from verified records. Joined roles such as 'Associate Professor+Postdoc' are split per affiliation, e-mail domains veto affiliation-name matches that point elsewhere, and later records reveal moves (31 self-reports excluded; 28 flagged where CSRankings disagrees).
- **Style dictionary.** 264 profiles (global, venue, topic and lab corpora) from 4013 accepted papers (691 with arXiv full text): sentence shape, stance (hedges, boosters, promotional terms), voice, citation and number density, rhetorical moves and generic formulas. A hashed n-gram index guards against verbatim reuse; no paper text is redistributed.
- **Rules.** 12 venue profiles (NeurIPS/ICML/ICLR/CVPR/ICCV/ECCV 2025-2027, TPAMI, Nature, Nature Machine Intelligence) and 148 atomic writing rules distilled from venue checklists, reviewer guides and public advice by Freeman, Black, Peyton Jones, Widom, Lipton & Steinhardt, Huang, Karpathy and others - each with sources.
- **Research map.** 240 curated accepted papers (including 38 award winners) and 172 GitHub repositories with API-verified metadata and licence caveats.
- **Software.** A FastAPI + React application (localhost) and a GitHub Pages build (browser mode). Tests: 33 passed, 0 failed. On 30 synthetic domain fixtures the mechanical checks detect 22/22 meaning-changing edits with 0/8 false alarms on safe edits (development fixtures, not a benchmark).

What is not claimed. No live model was available in the build environment (no API key), so the quality of model-written drafts and model critiques has not been measured; only the deterministic layer, the workflow and the mock path were executed. A style profile describes a paper collection (co-authored, edited text), not an individual's mind. Inclusion in a list is evidence of publication breadth, not a ranking of scientific merit.

2 Scope, definitions and evidence rules

Rule	Operational meaning
Institution filter	QS World University Rankings 2027 overall rank <= 100, ties included (102 institutions). The rank attaches to the professor's current institution.
List A	At least one accepted main-track paper at each of NeurIPS, ICML, ICLR, CVPR and ICCV.
List B	List A plus at least one accepted main-track ECCV paper (List B is a subset of List A).
Tracks	Main proceedings; NeurIPS Datasets & Benchmarks and ICML/NeurIPS position papers are labelled. Workshops, journal-to-conference presentations, rejected, withdrawn and conditional decisions never count.
Venue presence	CSRankings DBLP-derived count > 0, or a strongly attributed accepted-paper witness, or a verified 2026-10-

Rule	Operational meaning
	26 dossier witness outside paper-list coverage.
Identity	Seed title, affiliation/e-mail domain, DBLP id, >= 2 shared co-authors, or exact rare name. A conflicting DBLP id always rejects a match. Common names need affiliation, DBLP id or co-author evidence.
Topic relevance	Weighted count of accepted six-venue papers whose own title/keywords match the ontology (narrow topics weight 1; 3D vision and safety 0.5; abstract-only matches half); threshold 3.
Evidence levels	V1 all required venues linked with strong identity; V2 some rest on counts/dossiers. R1 CSRankings affiliation + OpenReview professor title in 2024+; R2 one role source. E2 = V1 or R1 (compatible with the 26 October rubric); E3 otherwise.
Professor	Assistant, associate or full professor (incl. research professor). Adjunct, visiting, emeritus, honorary and courtesy titles are excluded from the OpenReview channel.

Nature-family and TPAMI papers enrich the research map (Section 5) but never fill a missing conference cell. Topic tags describe published output, not expertise; a famous co-author is not evidence of who wrote a paper's prose.

3 How the professor census works

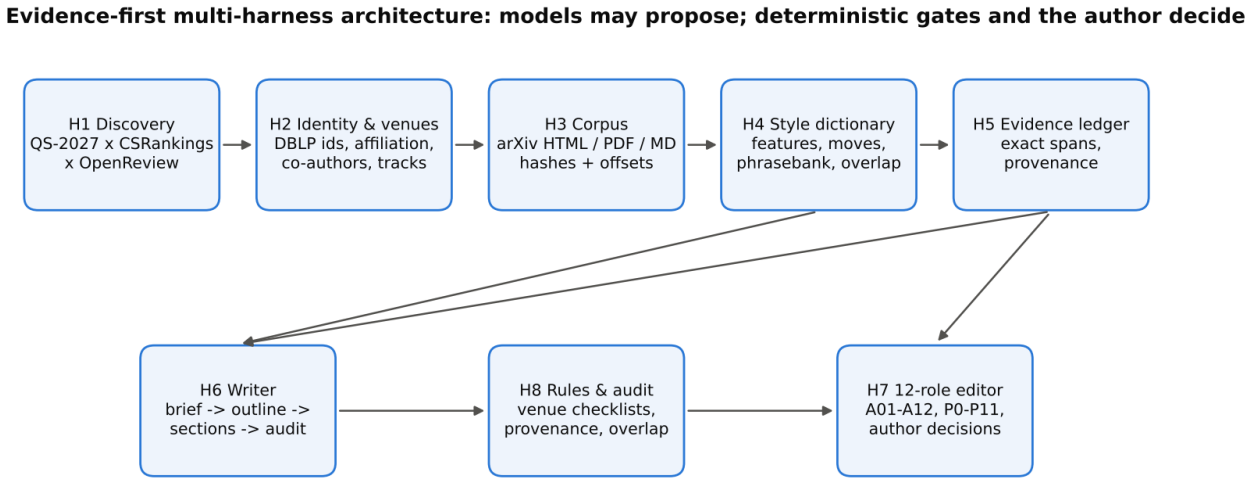


Figure 1. Harness architecture. H1-H2 produce the professor lists in this report.

3.1 Three discovery channels

Channel	What it contributes	Main limitation
CSRankings faculty file + generated-author-info.csv	Current affiliation, homepage and DBLP-derived main-track paper counts per conference and year for 3,571 QS-institution faculty with six-venue papers.	Covers faculty who can advise CS PhD students; misses some EE/ME/BCS faculty and people on leave or in industry roles; DBLP lags the newest proceedings.
OpenReview self-reported professors (Paper Copilot lists)	Authors of accepted ICLR/NeurIPS/ICML papers whose profile states a professor title at a QS top-100 institution (>= 2 verified records, latest in 2023 or later).	Self-reported; only people active at OpenReview venues; titles can be stale.
Seed dossiers (2026-10-26 report)	162 verified witness titles for F01-F27 anchor identities, including movers (e.g. Hao Su to Fudan) and faculty outside CSRankings (e.g. Tenenbaum, Vedaldi, Zisserman).	Only as complete as the earlier audit.

3.2 Identity resolution and venue verification

- Venue counts come from CSRankings' DBLP identities (disambiguated names such as 'Xiaolong Wang 0004').

- 94,806 accepted papers from Paper Copilot (NeurIPS 2010-2025, ICML 2017-2026, ICLR 2013-2026, CVPR 2017-2025, ICCV 2017-2025, ECCV 2018-2024, CoRL, RSS) supply witness titles and primary links; rejected/withdrawn/workshop records are excluded by status and track.
- Matching passes: seed titles, affiliation strings and e-mail/affiliation domains (102 curated institution patterns with exclusions such as CUHK-Shenzhen vs CUHK, HKUST-Guangzhou vs HKUST, British Columbia vs Columbia), DBLP-id learning from verified records, co-author continuity, and exact rare names.
- Profile-field permutation detection: a homepage in a slot that names a different co-author marks the record misaligned; its per-author fields are discarded rather than guessed.
- Self-reported roles: '+'-joined titles are paired with their own affiliations (a postdoc at a QS university with a professorship elsewhere is not counted); an institutional e-mail domain that belongs to another institution vetoes an affiliation-name match (HKUST-Guangzhou, CUHK-Shenzhen, Polytechnique Montreal, NTHU vs Tsinghua, UCLouvain vs UCL); a later OpenReview record with another role marks a move. Where CSRankings names a different institution the entry is kept with an affiliation note to verify (data/faculty/self_report_conflicts.csv).
- Presence also accepts strongly attributed witnesses for proceedings DBLP has not indexed yet (e.g. ICML 2026).

3.3 Reconciliation with the 26 October lists

Earlier source	Status now	Count
discovery queue	confirmed List A	7
discovery queue	confirmed List B	50
discovery queue	not found at a QS top-100 institution in any channel	33
discovery queue	not on List A	72
report F01-F27 (six venues)	confirmed List B	27
report specialists (not certified)	confirmed List A	1
report specialists (not certified)	confirmed List B	4
report specialists (not certified)	not found at a QS top-100 institution in any channel	2
report specialists (not certified)	not on List A	8
strict CSV (List A only)	confirmed List A	3
strict CSV (List B)	confirmed List B	30

Typical reasons an earlier name is not on List A now: no main-track paper at one venue in the evidence (e.g. a robotics specialist without ICCV papers), or a different person with the same name. These are not judgements of quality. The full row-level comparison is in data/faculty/prior_list_reconciliation.csv.

4 Results: List A and List B in numbers

Source	List A	of which List B	Topic-relevant List B
csrankings	589	476	362
openreview_self_reported	141	97	74
seed_dossier	8	8	8
Total	738	581	444

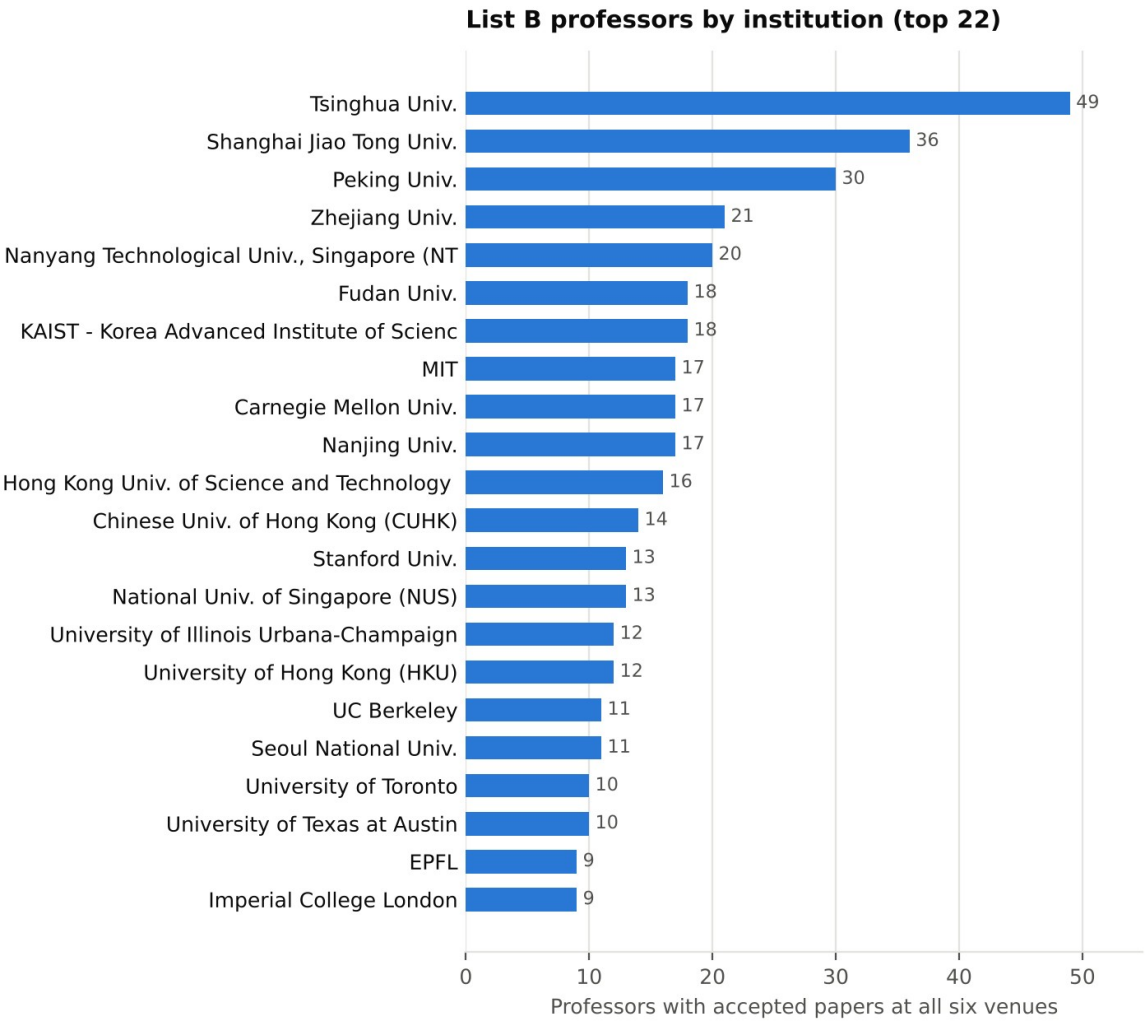


Figure 2. List B professors by institution (top 22).

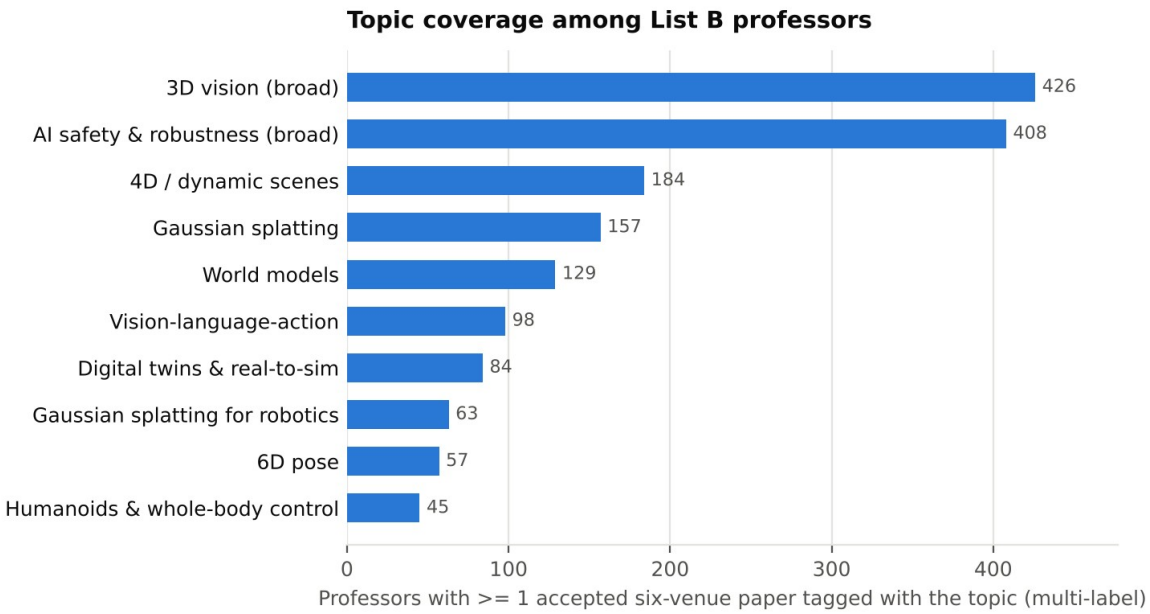


Figure 3. Topic coverage among List B professors (multi-label).

The per-professor tables of the full edition (top 40 List B professors by topic score, the leaders of each topic and strong topic output outside the QS top 100) are not part of this public edition.

5 Research landscape and open problems

The organising principle is the loop from observation to representation, predicted consequences, selected action and measured outcome. A scene representation, a video predictor, a control policy and a calibrated simulator occupy different parts of that loop; calling all of them 'world models' hides what is actually evaluated. For each direction below: what the field is converging on, the distinctions a paper must keep, and open questions worth a paper. The tables list curated accepted papers (venue, year, award where applicable) from the research map; confidence labels are in data/papers/top_papers.json.

5.1 World models

Two lines are converging: control-oriented latent world models used for planning and policy learning (the Dreamer and TD-MPC families) and generative interactive video models used as simulators and data engines (Genie, UniSim, navigation and long-memory video world models). Action conditioning, memory and 3D consistency over long horizons are now the frontier, together with learning from video without action labels.

Distinctions a paper must keep.

- Prediction quality (FVD/PSNR), action-conditioned accuracy and closed-loop control utility are three different claims; report them separately.
- A latent action inferred from video is not an actuator command; a plausible rollout is not a calibrated physical simulator.
- State the horizon, the training distribution and which embodiments were held out.

Open questions. Action-space alignment across embodiments; Calibrated uncertainty over long rollouts; Evaluation protocols that link prediction errors to decision quality; Which details a predictive model can discard while preserving decisions.

Paper	Venue	Year	Award / status
NitroGen: An Open Foundation Model for Generalist Gaming Agents	CVPR	2026	CVPR 2026 Best Paper Honorable Mention
Navigation World Models	CVPR	2025	CVPR 2025 Best Paper Honorable Mention
Genie: Generative Interactive Environments	ICML	2024	ICML 2024 Best Paper Award
Learning Interactive Real-World Simulators	ICLR	2024	ICLR 2024 Outstanding Paper Award
Robust agents learn causal world models	ICLR	2024	ICLR 2024 Outstanding Paper Honorable Mention
Generative Image Dynamics	CVPR	2024	CVPR 2024 Best Paper Award
Latent Particle World Models: Self-supervised Object-centric Stochastic Dynamics Modeling	ICLR	2026	Oral
WoW!: World Models in a Closed-Loop World	ICLR	2026	Oral
Ctrl-World: A Controllable Generative World Model for Robot Manipulation	ICLR	2026	Poster

5.2 Vision-language-action

Vision-language-action models map images and instructions to robot actions; the design space now centres on action representation (discrete tokens vs continuous diffusion/flow heads, chunking), cross-embodiment data mixtures, reasoning traces grounded in actions, efficient fine-tuning and distributed real-world evaluation.

Distinctions a paper must keep.

- Policy frequency is not controller frequency.
- Separate seen tasks, novel semantics and novel motor skills; report trials per task and the success definition.
- Simulation success is not real-robot success; generated explanations are outputs, not faithful accounts of computation.

Open questions. Data-efficient adaptation to new embodiments; Safety constraints and failure detection inside VLA policies; Latency-aware action decoding; Standardised real-world evaluation.

Paper	Venue	Year	Award / status

NitroGen: An Open Foundation Model for Generalist Gaming Agents	CVPR	2026	CVPR 2026 Best Paper Honorable Mention
Learning Interactive Real-World Simulators	ICLR	2024	ICLR 2024 Outstanding Paper Award
WoW!: World Models in a Closed-Loop World	ICLR	2026	Oral
Ctrl-World: A Controllable Generative World Model for Robot Manipulation	ICLR	2026	Poster
DreamDojo: A Generalist Robot World Model from Large-Scale Human Videos	ICML	2026	Spotlight
SimpleVLA-RL: Scaling VLA Training via Reinforcement Learning	ICLR	2026	Poster
Cosmos Policy: Fine-Tuning Video Models for Visuomotor Control and Planning	ICLR	2026	Poster
WholeBodyVLA: Towards Unified Latent VLA for Whole-body Loco-manipulation Control	ICLR	2026	Poster
RoboCasa365: A Large-Scale Simulation Framework for Training and Benchmarking Generalist Robots	ICLR	2026	Poster

5.3 Gaussian splatting

Explicit Gaussian scene representations moved from offline novel-view synthesis to SLAM (dense RGB-D, monocular and dynamic-scene variants), feed-forward reconstruction, anti-aliasing and robot-centric uses: feature fields for manipulation, physics-augmented Gaussians and real-to-sim scene construction.

Distinctions a paper must keep.

- Rendering speed is not end-to-end SLAM throughput; report tracking (ATE) and mapping quality separately.
- State sensor assumptions (monocular, RGB-D, stereo), metric scale and static/dynamic scene assumptions.
- Photometric quality is not geometric accuracy, and neither is manipulation success.

Open questions. Dynamic and articulated scenes with uncertainty; Physical property estimation for simulation; Memory-bounded large-scale maps; Closing the loop from map to control.

Paper	Venue	Year	Award / status
3D Student Splatting and Scooping	CVPR	2025	CVPR 2025 Best Paper Honorable Mention
pixelSplat: 3D Gaussian Splats from Image Pairs for Scalable Generalizable 3D Reconstruction	CVPR	2024	CVPR 2024 Best Paper Honorable Mention
Mip-Splatting: Alias-free 3D Gaussian Splatting	CVPR	2024	CVPR 2024 Best Student Paper
LSD-SLAM: Large-Scale Direct Monocular SLAM	ECCV	2014	ECCV 2024 Koenderink Prize (Test of Time)
SoMA: A Real-to-Sim Neural Simulator for Robotic Soft-Body Manipulation	ICML	2026	Poster
D-REX: Differentiable Real-to-Sim-to-Real Engine for Learning Dexterous Grasping	ICLR	2026	Poster
Dream to Manipulate: Compositional World Models Empowering Robot Imitation Learning with Imagin	ICLR	2025	Poster
Building Interactable Replicas of Complex Articulated Objects via Gaussian Splatting	ICLR	2025	Poster
WildGS-SLAM: Monocular Gaussian Splatting SLAM in Dynamic Environments	CVPR	2025	Accepted (paperlists label: Poster; oral/highlig

5.4 Humanoids & whole-body control

Learning-based whole-body control has advanced through large-scale simulation, motion imitation from human data and explicit sim-to-real alignment of physics; universal controllers expose several command modes (navigation, teleoperation, manipulation) through one policy.

Distinctions a paper must keep.

- Keep simulated and physical results separate; give trial counts, falls and recovery behaviour.
- Tracking error is not task completion.
- State contact assumptions, disturbances and hardware.

Open questions. Contact-rich loco-manipulation; Scaling with human video and teleoperation data; Safety, fall avoidance and recovery; Comparable evaluation across hardware platforms.

Paper	Venue	Year	Award / status
BFM-Zero: A Promptable Behavioral Foundation Model for Humanoid Control Using	ICLR	2026	Poster

Paper	Venue	Year	Award / status
Unsupervised Rein			
WholeBodyVLA: Towards Unified Latent VLA for Whole-body Loco-manipulation Control	ICLR	2026	Poster
PlayerOne: Egocentric World Simulator	NeurIPS	2025	Oral
Whole-Body Conditioned Egocentric Video Prediction	NeurIPS	2025	Poster
Hierarchical World Models as Visual Whole-Body Humanoid Controllers	ICLR	2025	Poster
Zero-Shot Whole-Body Humanoid Control via Behavioral Foundation Models	ICLR	2025	Poster
InterMimic: Towards Universal Whole-Body Control for Physics-Based Human-Object Interactions	CVPR	2025	Highlight
KungfuBot: Physics-Based Humanoid Whole-Body Control for Learning Highly-Dynamic Skills	NeurIPS	2025	Poster
From Experts to a Generalist: Toward General Whole-Body Control for Humanoid Robots	NeurIPS	2025	Spotlight

5.5 Digital twins & real-to-sim

Real-to-sim pipelines build simulated counterparts of real scenes (digital twins and 'digital cousins' that vary scenes while preserving affordances), procedural environment generation scales embodied training, and learned interactive simulators complement physics engines.

Distinctions a paper must keep.

- A static reconstruction is a replica; a twin needs synchronisation and validated conditions.
- Visual fidelity and physical validity are separate axes; report calibration and validation tasks.
- Keep sim-to-real success tied to the task conditions it was measured under.

Open questions. Automatic inference of physical parameters; Uncertainty-aware transfer; Does more realism improve held-out task performance?; Continual synchronisation.

Paper	Venue	Year	Award / status
SAM 3D: 3Dfy Anything in Images	CVPR	2026	CVPR 2026 Best Paper Honorable Mention
Generating Physically Stable and Buildable Brick Structures from Text	ICCV	2025	ICCV 2025 Best Paper Award (Marr Prize)
Learning Interactive Real-World Simulators	ICLR	2024	ICLR 2024 Outstanding Paper Award
ProcTHOR: Large-Scale Embodied AI Using Procedural Generation	NeurIPS	2022	NeurIPS 2022 Outstanding Paper Award
LSRM: High-Fidelity Object-Centric Reconstruction via Scaled Context Windows	ECCV	2026	ECCV 2026 Best Paper Honorable Mention
Do Differentiable Simulators Give Better Policy Gradients?	ICML	2022	ICML 2022 Outstanding Paper Award
SoMA: A Real-to-Sim Neural Simulator for Robotic Soft-Body Manipulation	ICML	2026	Poster
RoboCasa365: A Large-Scale Simulation Framework for Training and Benchmarking Generalist Robots	ICLR	2026	Poster
RobotArena ∞: Unlimited Robot Benchmarking via Real-to-Sim Translation	ICLR	2026	Poster

5.6 AI safety & robustness

Technical safety spans alignment training (RLHF to direct preference optimisation), robustness to jailbreaks and adaptive attacks, evaluation suites, interpretability and scalable oversight (e.g. debate), plus safe control for embodied systems. Award papers in 2024-2026 stress that shallow alignment is brittle and that evaluation must be adversarial and representative.

Distinctions a paper must keep.

- A finite test with zero failures is not a guarantee; report the exposure denominator.
- Preference, factuality and physical safety are different objectives.
- State the threat model and whether attacks were adaptive.

Open questions. Robustness under adaptive adversaries; Oversight that scales with capability; Safety of embodied agents under distribution shift; Interpretability that changes decisions.

Paper	Venue	Year	Award / status
Safety Alignment Should be Made More Than Just a Few Tokens Deep	ICLR	2025	ICLR 2025 Outstanding Paper Award
Robust agents learn causal world models	ICLR	2024	ICLR 2024 Outstanding Paper Honorable Mention
Debating with More Persuasive LLMs Leads to More Truthful Answers	ICML	2024	ICML 2024 Best Paper Award
Stealing part of a production language model	ICML	2024	ICML 2024 Best Paper Award
The PRISM Alignment Dataset: What Participatory, Representative and Individualised Human Feedba	NeurIPS	2024	NeurIPS 2024 Best Paper (Datasets & Benchmarks t
Concept Arithmetics for Circumventing Concept Inhibition in Diffusion Models	ECCV	2024	ECCV 2024 Best Paper Honorable Mention
Direct Preference Optimization: Your Language Model is Secretly a Reward Model	NeurIPS	2023	NeurIPS 2023 Outstanding Main Track Runner-Up
DecodingTrust: A Comprehensive Assessment of Trustworthiness in GPT Models	NeurIPS	2023	NeurIPS 2023 Outstanding Paper (Datasets & Bench
Intriguing properties of neural networks	ICLR	2014	ICLR 2024 Test of Time Award (runner-up)

5.7 3D vision

Feed-forward geometry models predict cameras, depth and point maps directly from images (the DUSt3R / VGGT line; VGGT received the CVPR 2025 Best Paper Award), alongside large reconstruction models and 3D generation with structured latents.

Distinctions a paper must keep.

- Scale ambiguity, camera assumptions and view coverage determine what a reconstruction proves.
- Evaluate on held-out scenes and report geometry, not only rendering.

Open questions. Metric-scale reconstruction in the wild; 3D foundation features for robotics; Uncertainty-aware geometry.

Paper	Venue	Year	Award / status
Native and Compact Structured Latents for 3D Generation	CVPR	2026	CVPR 2026 Best Student Paper Award
SAM 3D: 3Dfy Anything in Images	CVPR	2026	CVPR 2026 Best Paper Honorable Mention
Efficiently Reconstructing Dynamic Scenes One D4RT at a Time	CVPR	2026	CVPR 2026 Best Paper Award
VGGT: Visual Geometry Grounded Transformer	CVPR	2025	CVPR 2025 Best Paper Award
3D Student Splatting and Scooping	CVPR	2025	CVPR 2025 Best Paper Honorable Mention
Neural Inverse Rendering from Propagating Light	CVPR	2025	CVPR 2025 Best Student Paper
Generating Physically Stable and Buildable Brick Structures from Text	ICCV	2025	ICCV 2025 Best Paper Award (Marr Prize)
MegaSaM: Accurate, Fast and Robust Structure and Motion from Casual Dynamic Videos	CVPR	2025	CVPR 2025 Best Paper Honorable Mention
pixelSplat: 3D Gaussian Splats from Image Pairs for Scalable Generalizable 3D Reconstruction	CVPR	2024	CVPR 2024 Best Paper Honorable Mention

5.8 4D / dynamic scenes

Dynamic reconstruction and tracking (tracking any point, dynamic scene reconstruction from casual video, 4D Gaussians) matured quickly; CVPR 2026's Best Paper addressed efficient dynamic-scene reconstruction.

Distinctions a paper must keep.

- Time-varying geometry is not action-conditioned dynamics.
- Separate camera motion from scene motion.
- Report temporal consistency, not only per-frame accuracy.

Open questions. Long-horizon 4D memory; Physically consistent dynamics; Linking 4D perception to control.

Paper	Venue	Year	Award / status
Efficiently Reconstructing Dynamic Scenes One D4RT at a Time	CVPR	2026	CVPR 2026 Best Paper Award

Paper	Venue	Year	Award / status
VGGT: Visual Geometry Grounded Transformer	CVPR	2025	CVPR 2025 Best Paper Award
MegaSaM: Accurate, Fast and Robust Structure and Motion from Casual Dynamic Videos	CVPR	2025	CVPR 2025 Best Paper Honorable Mention
Generative Image Dynamics	CVPR	2024	CVPR 2024 Best Paper Award
DynIBaR: Neural Dynamic Image-Based Rendering	CVPR	2023	CVPR 2023 Best Paper Honorable Mention
Tracking Everything Everywhere All at Once	ICCV	2023	ICCV 2023 Best Student Paper
Depth Anything 3: Recovering the Visual Space from Any Views	ICLR	2026	Oral
Trellis4D: Complex 4D Mesh Generation	NeurIPS	2026	Oral
Aether: Geometric-Aware Unified World Modeling	ICCV	2025	Accepted (paperlists label: Poster; oral/highlig

5.9 6D pose

Model-based and model-free novel-object pose estimation and tracking (FoundationPose), category-level pose (normalised object coordinates), and probabilistic PnP layers define the area; BOP-style evaluation is standard.

Distinctions a paper must keep.

- State the transform direction and coordinate frames.
- Handle symmetry explicitly (ADD vs ADD-S) and report thresholds.
- CAD-based and reference-image settings are different problems.

Open questions. Pose under heavy occlusion and transparency; Pose for articulated and deformable objects; Uncertainty-aware pose for grasping.

Paper	Venue	Year	Award / status
EPro-PnP: Generalized End-to-End Probabilistic Perspective-N-Points for Monocular Object Pose E	CVPR	2022	CVPR 2022 Best Student Paper
6D Object Pose Tracking in Internet Videos for Robotic Manipulation	ICLR	2025	Poster
Any6D: Model-free 6D Pose Estimation of Novel Objects	CVPR	2025	Accepted (paperlists label: Poster; oral/highlig
CRISP: Object Pose and Shape Estimation with Test-Time Adaptation	CVPR	2025	Highlight
6DOPE-GS: Online 6D Object Pose Estimation using Gaussian Splatting	ICCV	2025	Accepted (paperlists label: Poster; oral/highlig
FoundationPose: Unified 6D Pose Estimation and Tracking of Novel Objects	CVPR	2024	Highlight
SAM-6D: Segment Anything Model Meets Zero-Shot 6D Object Pose Estimation	CVPR	2024	Accepted (paperlists label: Poster; oral/highlig
GigaPose: Fast and Robust Novel Object Pose Estimation via One Correspondence	CVPR	2024	Accepted (paperlists label: Poster; oral/highlig
HouseCat6D - A Large-Scale Multi-Modal Category Level 6D Object Perception Dataset with Househo	CVPR	2024	Highlight

6 The style dictionary: how these labs write

The style dictionary answers 'what does a successful paper in this area typically look like, section by section?' with measurements, so that writing guidance is evidence-based and never a matter of copying. It is built from the accepted papers attributed to the topic-relevant professors in Section 4.

Corpus measure	Value
Documents (accepted papers)	4013
With arXiv full text (sections)	691
Lab corpora (professors)	246
Profiles (global + venue + topic + lab)	264
Phrasebank formulas	381
Overlap-guard hashes	892,005

6.1 What is measured

Feature family	Measurements	How the writer uses it
Sentence shape	mean / SD / p90 sentence length, MATTR lexical diversity, long-word rate	match the compression level typical of the venue and section
Stance (metadiscourse)	hedges, boosters, promotional terms, self-mention ('we/our'), engagement per 1k words	set claim strength: hedge interpretations, state measurements plainly, avoid hype
Voice and nominalisation	passive constructions per sentence, nominalisation rate	keep actions in verbs and actors visible
Evidence density	citations, numbers and inline math per sentence	e.g. related work vs experiments
Rhetorical moves	abstract: background, gap, purpose, method, result, implication, resource; introduction: territory, niche, occupy (Swales' CARS)	plan move order; contributions list placement
Formulas	content-masked n-grams recurring across >= 4 labs (generic by construction)	optional connective phrasing; never content
Function words	relative frequencies of 150 function words (Burrows' Delta)	measure style distance of a draft

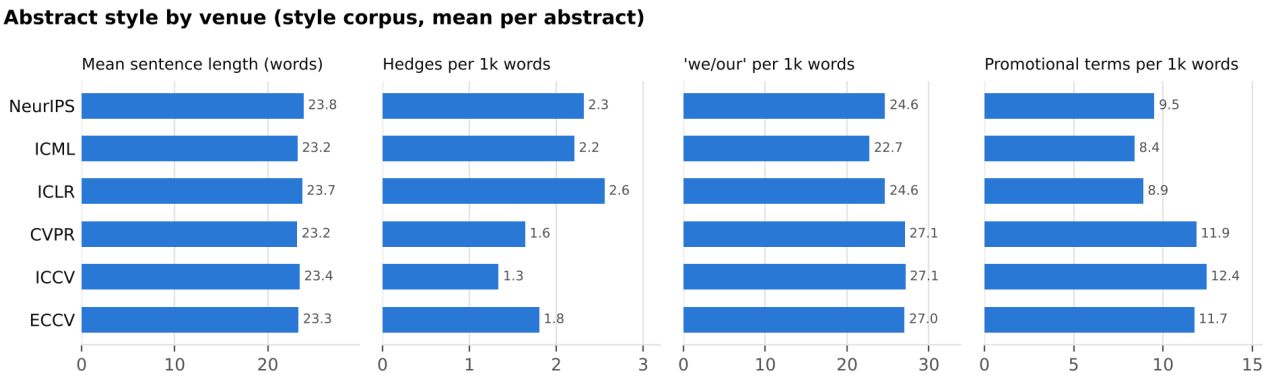


Figure 4. Abstract-level style by venue in the style corpus.

Reading the figure. Abstracts at the vision venues hedge less (1.6 vs 2.4 hedges per 1k words at NeurIPS/ICML/ICLR), use more promotional terms (12.0 vs 8.9) and more 'we/our' (27.1 vs 24.0), with similar sentence length (about 23 words). These are descriptive statistics of accepted papers whose topic mix differs by venue, not evidence that a style causes acceptance; the writer uses them as ranges, and the rule set still discourages unsupported promotional claims.

6.2 Guidance derived from the whole corpus

- Abstracts: about 180 words in 7.9 sentences (mean sentence 23.5 words, IQR 20.7-25.6).
- Common abstract move sequences: B-G-P-M-R (4%), P-M-R (3%), B-M-P-M-R (2%) (B=background, G=gap, P=purpose, M=method, R=result, I=implication, S=resource).
- Abstract stance: hedges 2.0/1k words, boosters 7.4/1k, promotional terms 10.5/1k, 'we/our' 25.6/1k.

- Preferred self-reporting verbs after 'we': propose, introduce, present, show, demonstrate, design.
- Introductions: about 737 words; mean sentence 24.8 words; citations per sentence 0.38.
- 68% of introductions state an explicit contribution list; the first 'occupy-the-niche' sentence appears on average at 37% of the introduction.
- Method sections: passive constructions 0.28/sentence, inline math 1.09/sentence, mean sentence 22.6 words.
- Experiment sections: numbers per sentence 0.74; hedges 3.2/1k words.

6.3 Generic formulas (phrasebank)

Content words are masked as [X]; a formula is kept only if it recurs in many papers from at least four labs, so these are shared conventions rather than anyone's distinctive prose.

Formula	Papers	Labs
we propose a	1126	235
in this paper, we	974	220
in this work, we	650	200
we introduce a	524	192
the effectiveness of	461	174
due to the	459	179
we show that	438	163
to address this	413	181
the [X] of the	405	186
we propose a novel	405	172
we propose a [X]	379	170
show that our	368	160
in this paper, we propose	345	151
as well as	342	167
that our method	340	148
demonstrate that our	336	161
code is available at	315	133
we present a	308	143
we propose [X], a	306	157
based on the	305	155
extensive experiments on	300	126
large language models	298	125
we introduce [X], a	287	147
experiments show that	280	142
to address these	268	145

Lab profiles (one per professor) are part of the full edition and the local app; this public edition describes them only in aggregate.

Ethics and attribution. Profiles are statistics of co-authored, edited text and are attributed to a collection. The writer receives numbers and generic formulas, never corpus sentences, and every draft is screened against the hashed n-gram index for verbatim overlap. Imitating a person's identity or distinctive passages is out of scope.

7 Venue requirements and writing rules

Venue	Main-text limit	Abstract	Anonymity
NeurIPS 2026	9 content pages (text, figures and tables all count)	Exactly one paragraph (template).	Double-blind. Compile in the default anonymous mode ('Anonymous Author
ICML 2026	8 pages of main body	Unverified for 2026.	Double-blind. Submissions must contain nothing intended to, or with th
ICLR 2026	9 pages of main text at initial submission	One paragraph (template).	Double-blind on OpenReview; non-anonymous submissions are rejected wit
ICLR 2027	9 pages of main text at initial submission	One paragraph (template).	Double-blind; revealing author identity in the main text or supplement

Venue	Main-text limit	Abstract	Anonymity
CVPR 2026	8 pages including figures and tables	No fixed limit; follow the style and length of previous	Double-blind. Cite own work in the third person; no acknowledgments in
CVPR 2027	8 pages including figures and tables	No fixed limit (template).	Double-blind, same rules as CVPR 2026 (template). The 2027 template re
ICCV 2025	8 pages including figures and tables	No fixed limit (CVPR/ICCV author kit).	Double-blind, same conventions as the CVPR/ICCV author kit.
ICCV 2027			
ECCV 2026	14 LNCS pages including figures and tables	150	Double-blind: reviewers/ACs must not be able to infer authors beyond r
IEEE TPAMI Regular paper (ro	12 double-column pages including references and author biogr	250	Review model (single- vs double-anonymous) not verified in this sessio
Nature Article (current auth	None words and display items	200	Not verified in this session (Nature has offered optional double-anony
Nature Machine Intelligence	3500 words of main text	150	Not verified in this session.

Unverified items are marked inside data/rules/writing_rulebook.json with the year that was confirmed. Always re-check the official call for papers before submission.

148 rules. 39 are checked by code today; 100 are reviewed by a model that must quote the manuscript verbatim (quotes are verified; this includes 43 rules designed for code checks that are not implemented yet); 9 are checklist items for the author. Categories: formatting (12), style (12), structure (11), experiments (10), introduction (9), claims (9), ethics (9), figures (8), anonymity (8), abstract (7), related_work (7), method (7), notation (7), reproducibility (7), rebuttal (7), tables (5), hedging (4), limitations (4), results (3), title (2).

Rule	What to do	Checked by	Source
R-ABS-001	Make the abstract a self-contained miniature of the paper that names the gap or problem, what you did, the main finding and why it matters, spending at most one	model	Jennifer Widom; Brett Mensh and Konrad Kording
R-ABS-003	Keep the abstract within the venue's length limit or recommendation.	engine	Nature Machine Intelligence; Nature (Springer Nature)
R-ABS-004	Write the abstract as a single paragraph.	engine	NeurIPS 2026 organizers; ICLR
R-ABS-005	Leave citations out of the abstract wherever the venue requires an unreferenced abstract.	engine	ECCV 2026 organizers; Nature Machine Intelligence
R-ABS-006	For a Nature Article, open with a fully referenced summary paragraph of roughly 200 words that a scientist outside the field can follow.	model	Nature (Springer Nature); Nature (Springer Nature)
R-INTRO-001	Make the introduction answer, in this order, what the problem is, why it matters, why it is hard (why obvious approaches fail), why earlier work falls short and	model	Jennifer Widom; Andrej Karpathy
R-INTRO-003	Phrase every contribution as a specific claim that the paper's evidence could refute, not as an activity.	model	Simon Peyton Jones
R-RW-002	For each topic, trace how the line of work developed and close by saying how your work relates to it, naming the key difference.	model	Jia-Bin Huang; NeurIPS 2026 organizers
R-RW-006	Cite all closely related and competing work, including recent work that weakens your novelty argument.	model	Jia-Bin Huang; NeurIPS 2026 organizers
R-METH-001	Justify every non-obvious design choice: why it is needed, what the alternatives are, and why this option won (ideally backed by an ablation).	model	Andrej Karpathy; Kayvon Fatahalian
R-METH-007	For each theoretical result, state all assumptions in or beside the statement, number and cross-reference it, and give a complete proof (with a short sketch in	model	NeurIPS 2026 organizers
R-EXP-001	Map every claim in the abstract and contribution list to at least one experiment, figure, table or theorem that tests it.	model	Jason Eisner; Andrej Karpathy
R-EXP-002	Include ablations that isolate the effect of each proposed component so the source of the improvement is clear.	model	Zachary C. Lipton and Jacob St
R-EXP-004	Report uncertainty (error bars, confidence intervals or significance tests) for the results behind the main claims, and say what variability it captures, how it	engine	NeurIPS 2026 organizers
R-EXP-005	Specify in the main text the setup needed to interpret the results (datasets and splits, preprocessing, exact metric protocols, key hyperparameters, optimizer,	model	NeurIPS 2026 organizers
R-EXP-006	Report compute: hardware type and memory, time or GPU-hours per run, total compute, and whether unreported preliminary or failed runs used substantially more.	engine	NeurIPS 2026 organizers
R-EXP-009	Compare against the strongest relevant and recent baselines under the same protocol, and never omit a known baseline because it performs well.	model	Andrej Karpathy; NeurIPS 2026 organizers
R-CLAIM-001	Make the claims in the abstract and introduction match what the results show, including their scope and key assumptions.	model	NeurIPS 2026 organizers; ICLR 2025 organizers
R-CLAIM-003	Support every explanation of why something works with evidence, or mark it	model	Zachary C. Lipton and Jacob

Rule	What to do	Checked by	Source
	explicitly as a hypothesis.		St
R-CLAIM-007	Scope generalization claims to what was actually evaluated (datasets, embodiments, environments, simulation vs. real world, number of runs).	model	NeurIPS 2026 organizers
R-CLAIM-008	Do not claim one method beats another when the gap lies within the reported uncertainty, and reserve 'significant(ly)' for differences backed by a statistical t	model	NeurIPS 2026 organizers
R-FIG-007	Place figure captions below figures and table captions above tables.	model	CVPR organizers (cvpr-org); ECCV 2026 organizers
R-FIG-008	Reference every figure and table in the text, near where it appears.	model	ECCV 2026 organizers
R-NOT-005	Define every symbol and acronym at or before its first use.	engine	Andrej Karpathy; Steven Pinker

8 Application architecture and the multiple harnesses

Harness	Input -> output	Gate that cannot be bypassed
H1 Discovery & scope	QS table, CSRankings, paper lists -> candidate people and papers	institution filter; accepted status and track
H2 Identity & venues	candidates -> List A / List B with witnesses	conflicting DBLP id rejects; misaligned metadata discarded
H3 Corpus	arXiv HTML / PDF / Markdown -> sectioned text with hashes and offsets	raw text stays local; parser warnings recorded
H4 Style dictionary	corpus -> profiles, phrasebank, overlap index	formulas need >= 4 labs; no sentences stored
H5 Evidence ledger	sources -> exact spans with provenance	quotes verified verbatim
H6 Writer	research brief -> outline -> sections -> audit -> revision	numbers only from the brief; citations only from the bibliography; placeholders instead of invention
H7 Scientific editor	Markdown -> candidates -> 12-role review -> author decision -> export	approval bound to candidate and context hashes; stale decisions rejected
H8 Rules & audit	draft -> rule findings, provenance, style distance, overlap	deterministic findings are reproducible; model rule reviews must quote

8.1 The 12 roles and the phases P0-P11

Phase	What happens	Roles
P0 Ingest	validate UTF-8 .md, keep original bytes, parse blocks, protect code/math/citations/URLs/tables	-
P1 Context	source-linked context record (definitions, quantities, constraints, unknowns) + author context	A07
P2 Draft	one candidate per paragraph (never writes the document)	A01
P3 Mechanical	protected content, numbers and bindings, citations, negation, modality, quantifiers, causality, risky domain substitutions	app
P4 Language	independent critics on the same candidate	A05, A06, A08, A09, A10
P5 Science	independent critics against original, candidate and context	A02, A03, A04, A07
P6 Rebuttal	uphold / challenge / duplicate / insufficient evidence for every finding	A11
P7 Revision	targeted repair of upheld material findings (bounded cycles)	A01
P8 Verification	fresh checks and critics on changed wording	A02-A10
P9 Judge	decision per finding; approve / request revision / needs author	A12
P10 Author	accept, reject, edit (re-verified), accept with explicit override	author
P11 Export	verified source-span patches, document-level checks, change report, audit JSON	app

- **Invariant:** an approval applies only to the exact candidate (SHA-256) and context revision reviewed; any new wording or glossary change invalidates it.
- **Majority never wins against evidence:** mandatory scientific reviews must complete; unresolved material/critical findings block automatic approval; author overrides are recorded and never labelled machine-verified; a no-change export is byte-identical to the upload.
- **Providers:** Claude API (default claude-opus-5-5 with adaptive thinking, explicit effort, structured JSON outputs, prompt caching and server-side refusal fallback), Claude on Microsoft Foundry (Azure billing; client-side fallback middleware), Azure OpenAI, OpenAI-compatible servers (e.g. self-hosted XtraGPT) and an offline mock. Each role can be routed to a different provider; the actual model is recorded per call.
- **Safety and privacy:** manuscript text is treated as data (prompt-injection resistant prompts), keys stay server-side and are masked, the server binds to localhost, uploads are size/encoding checked, previews are sanitised, and documents can be deleted.

9 GitHub repository map

172 repositories with metadata copied from the GitHub API on 2026-11-05: prose linting stylometry (17), research writing agents (16), document parsing scholarly (15), factuality citation eval (15), orchestration (15), scientific discourse revision (12), topic humanoids (11), topic 3d 4d 6d vision (10), topic world models (9), topic vla (9), peer review rebuttal (8), topic gaussian splatting robotics (8), topic digital twins sim (8), topic ai safety (8),

scholarly discovery (7), curated lists (4). Integration decisions: research_map_only (64), design_reference (57), optional_adapter (29), adopt_as_dependency (22).

Repository	Stars	Licence	What we borrow
stanford-oval/storm	31,786	MIT	Split pre-writing from writing: (1) perspective-guided question asking to enumerate what each section must cover, (2) an
Future-House/paper-qa	9,546	Apache-2.0	Use its 'gather evidence -> summarize each chunk relative to the claim with a score -> answer only from top-scored evide
SakanaAI/AI-Scientist	14,947	NOASSERTION	Section-by-section LaTeX drafting with a fixed number of self-reflection rounds per section, a compile-and-repair loop f
SakanaAI/AI-Scientist-v2	7,443	NOASSERTION	Its best-first tree search over alternatives (parameters in bfts_config.yaml) suggests keeping several candidate drafts
SamuelSchmidgall/AgentLaboratory	5,919	MIT	Explicit phases (literature review, experimentation, report writing) with co-pilot checkpoints where a human steers each
HKUDS/AI-Researcher	5,867	-	Their benchmark idea of grading generated work against reference papers suggests an evaluation for us: redraft a section
Technion-Kishony-lab/data-to-paper	836	MIT	Make data-chaining a hard rule: every number, comparison and percentage in a draft must link to an evidence record (resu
AkariAsai/OpenScholar	1,683	Apache-2.0	The self-feedback loop (draft -> feedback -> extra retrieval -> revision -> final citation-verification pass that drops
assafelovic/gpt-researcher	30,490	Apache-2.0	Planner -> parallel researchers -> report writer layout with source tracking per paragraph, and a 'local documents only'
TIGER-AI-Lab/ScholarCopilot	253	MIT	Treat 'citation needed here' as an explicit decision: the writer emits a placeholder that must be resolved against our c
Xtra-Computing/XtraGPT	5	-	Represent every edit request as (selected span, full-paper context, named criterion) and keep a per-section criteria lis
AutoSurveys/AutoSurvey	478	-	Parallel section drafting from one shared outline followed by a global integration pass; its citation-quality evaluation
IAAR-Shanghai/SurveyX	996	-	The requirement that every reference be pre-converted to Markdown and pre-processed before generation is the same corpus
allenai/ai2-scholarqa-lib	288	Apache-2.0	Quote-first synthesis: the model must extract verbatim supporting quotes before writing, and each section is generated o
THUDM/LongWriter	1,881	Apache-2.0	Plan with explicit per-paragraph word budgets so page limits (e.g. 9-page NeurIPS/ICLR main text) are enforced at planni
PaperDebugger/paperdebugger	1,554	AGPL-3.0	Edits delivered as reviewable hunks anchored to positions in the LaTeX source (accept/reject one at a time) and critics
zhu-minjun/Researcher	409	NOASSERTION	Structure the judge like DeepReviewer: retrieval-backed novelty check -> per-dimension assessment -> reliability/self-ve
ResearAI/DeepReviewer-v2	596	MIT	A reviewer that must ground each weakness in a retrieved source or a paper span, behind a clean parser -> agent -> repor
allenai/marg-reviewer	66	Apache-2.0	Chunk-owning workers plus a coordinating leader lets critics reason over a whole paper beyond one context window, and th
Ahren09/AgentReview	701	Apache-2.0	Reviewer persona knobs (expert, lazy, adversarial) to stress-test our rebuttal agent, and the staged rebuttal -> discuss
maxidl/openreviewer	18	-	A cheap, specialised 'hostile reviewer' to generate likely objections for the rebuttal agent to pre-empt, using the offi
allenai/PeerRead	432	-	Aspect-score schema (e.g. clarity, originality, soundness, substance, impact) as a starting rubric for the judge, plus d
nnkennard/DISAPERE	1	NOASSERTION	Make the rebuttal agent's output an explicit alignment: every reviewer point maps to a response sentence with an action
UKPLab/nlpeer	36	Apache-2.0	Its data model linking paper versions, reviews and revisions is a good schema for our corpus, so reviewer comments can b
Leey21/awesome-ai-research-writing	35,610	-	Mine it for critic personas and checklists (e.g. a reviewer persona that tolerates no logic gaps) and for how writing sk
handsome-rich/Awesome-Auto-Research-Tools	1,273	CC0-1.0	Use as a watchlist feed; the LLM-API column quickly shows which systems already support Azure OpenAI or Anthropic models
natnew/awesome-ai-scientists	20	MIT	Source of benchmarks for scientific critique and overclaim detection to evaluate our evidence and verb/modality critics.
ai4s-research/awesome-ai-for-science	2,074	MIT	Pointers to paper-search agents and MCP servers that could feed the corpus builder, and to figure/chart tools for the fi
grobidOrg/grobid	5,218	Apache-2.0	Make it the canonical PDF -> structure step of the corpus builder: TEI gives paragraph and section IDs, in-text citation
grobidOrg/grobid-client-python	416	Apache-2.0	Use for corpus ingestion with concurrency control and resumable batches; keep raw TEI next to derived JSON for provenanc

Licence caveats. Several widely used research assets are non-commercial or copyleft (e.g. the original 3D Gaussian Splatting code, DUSt3R/MASt3R, FoundationPose, CoTracker; AGPL PyMuPDF; GPL tools) and the CSRankings data is CC

BY-NC-ND. Repository popularity is not evidence of correctness or reproducibility; check licences before reuse or redistribution. Details per repository are in data/repos/github_repos.json.

10 Verification, evaluation and release gates

What was verified	Result
Automated test suite (parser, patches, checks, rules, writer, editor workflow, API)	33 passed, 0 failed
Domain fixtures: harmful edits detected by mechanical checks	22/22
Domain fixtures: safe edits raising material/critical findings	0/8
No-change export byte-identical (LF, CRLF, BOM)	verified in tests
Stale / overlapping patches and stale author decisions rejected	verified in tests (HTTP 409)
Harmful writer edit cannot be auto-approved; override recorded as unverified	verified in tests
Missing provider leaves reviews 'skipped_unavailable', never 'passed'	verified in tests
Live model quality (drafts, critiques, Judge)	NOT measured - no API key in the build environment

Fixture results by expected check: causality 1/1, certainty 1/1, claim_strength 1/1, negation 1/1, numbers 2/2, numeric_binding 2/2, protected_content 2/2, quantifier 1/1, risky_substitution 10/10, statistics 1/1, terminology 1/1. These fixtures were written alongside the checks; treat them as regression tests, not as evidence of general accuracy.

10.1 Evaluation plan for model-assisted editing

- Datasets: ParaReval (instruction following, acceptability, preference; keep its splits) and HPMET's arXiv meaning-preservation subset (pairwise a/b/tie; judgements are not unique sentences) - loaders are included; data is not bundled.
- Expert fixtures: extend the 30 synthetic fixtures with author-reviewed paragraphs from your own drafts, split by source paper.
- Ablations under matched budgets: Writer only; Writer + one scientific critic; deterministic checks + Writer; 9 roles without the POS-style critics; full 12 roles. Report harmful-edit acceptance, safe-edit rejection, issue localisation, abstention, author effort, latency, tokens and cost by error category, with intervals.

Release gate	Condition
Exact extraction	100% of exported quotations match the canonical source span and version
Number and citation provenance	0 unsupported numbers and 0 unknown citation keys in a draft marked final
Meaning preservation	no known critical fixture auto-approved; material findings require explicit author override
Professor lists	every list entry has role evidence and one witness per required venue; conflicts surfaced
Reproducibility	repeated runs keep inputs, prompt hashes, model ids and configuration in the audit

11 Running the application and writing your paper

11.1 Localhost (full application)

- Windows: double-click launch_app.cmd (or run launch_app.ps1). macOS/Linux: ./launch_app.sh. The script creates a virtual environment, installs the package and opens http://127.0.0.1:8765.
- Manual: python -m venv .venv; activate; pip install -e .; python -m extractive_harness serve --open.
- Keys: put them in .env (copy .env.example) or enter them on the Settings page; they are stored only in .harness/config.json on your machine and are never returned by the API.

Option	What to configure	Billing
Claude API	ANTHROPIC_API_KEY; model claude-opus-5-5 (default), claude-sonnet-5-5 or claude-haiku-5-5; effort	Anthropic account (your Claude credits)
Claude on Microsoft Foundry (Azure)	ANTHROPIC_FOUNDRY_API_KEY, Foundry resource name, deployment name	Microsoft Marketplace through Azure; check whether your Azure credits apply to Marketplace models
Azure OpenAI	AZURE_OPENAI_ENDPOINT, AZURE_OPENAI_API_KEY, deployment, API version	Azure subscription credits
OpenAI-compatible	base URL + model (e.g. a vLLM server hosting XtraGPT)	your own hardware
Offline mock	tick 'template mode' / 'use offline mock backend'	free; labelled, not model review

Price basis used for estimates (USD per million input/output tokens, first-party rates that also apply to Claude on Foundry): Opus 5.5 $4 / $20, Sonnet 5.5 $2 / $10, Haiku 5.5 $0.10 / $0.50. Each editor paragraph makes about 11-14 role calls; set a run budget on the Run panel.

11.2 Public website (browser mode)

The public website (https://manish-genai.github.io/paper-writing-site/) is a static build of the same interface without per-professor data: institution, topic, paper, repository, rule and style explorers work without a server, and a browser-only Writer and Editor Lite call the Claude API directly with your key (kept in the browser tab unless you choose to remember it). The professor lists, lab style profiles and the full 12-role workflow are in the local application (Section 11.1).

11.3 Worked example: a made-up demo brief

The Writer's example brief describes a fictional project, object-centric Gaussian world models for robot manipulation (object-grouped 3D Gaussians, action-conditioned group dynamics, render-and-compare planning). It is not a real paper or result; its citations are real papers, and all result values are empty on purpose.

1. Open Writer, load the example brief, choose the venue (e.g. NeurIPS) and style profiles (venue-neurips plus topic-world-models), and run. Every missing number appears as [[RESULT: ...]].
2. Read the audit: rule findings (e.g. missing limitations or compute report), unsupported numbers, unknown citations, style distance, verbatim-overlap hits. Fill the brief's results table after experiments; re-run.
3. 'Send to Editor': build context (add protected terms such as method names and symbols), run the 12 roles, review each paragraph's diff and findings, accept or override explicitly, then compose and export Markdown, the change report and the audit JSON. LaTeX and BibTeX come from the Writer.
4. Before submission: run 'Check my manuscript' for the venue, complete the venue checklist, and verify every related-work claim against the cited paper (the brief's summaries are paraphrases).

Timing note. The rulebook records a CVPR 2027 paper deadline of CFP published. Submission deadline 2026-11-16 AoE; supplementary material until 2026-11-23 AoE. Verify dates on the official site.

12 Limitations and next steps

- The census denominator is CSRankings plus OpenReview self-reports plus earlier dossiers; robotics faculty in engineering departments who rarely publish at OpenReview venues may be missing, and self-reported titles can be stale. Next: verify roles against official faculty pages on your machine before outreach.
- QS ranks were reconstructed from QS and university announcements (69 high-, 33 medium-confidence entries).
- Paper lists end at NeurIPS 2025, ICML 2026, ICLR 2026, CVPR 2025, ICCV 2025 and ECCV 2024; later proceedings rely on CSRankings counts.
- Style profiles use abstracts for all labs and full text for a subset; section detection depends on LaTeXML headings. More full text per lab will make lab profiles more stable.
- Model-assisted quality is unmeasured until keys are configured; run the evaluation plan in Section 10.1 before trusting automatic approvals.
- Next research step: a focused corpus on one question (e.g. action-conditioned world models for cross-embodiment manipulation), with author-annotated paragraphs as a held-out set.

Appendix A QS World University Rankings 2027 top 100

Rank	Institution	Country	Confidence	List A	List B
1	Massachusetts Institute of Technology (MIT)	United States	high	20	17
=2	Imperial College London	United Kingdom	high	9	9
=2	Stanford University	United States	high	18	13
4	University of Oxford	United Kingdom	high	8	7
5	Harvard University	United States	high	2	1
6	University of Cambridge	United Kingdom	high	3	3
7	California Institute of Technology (Caltech)	United States	high	3	2
=8	ETH Zurich	Switzerland	high	9	7
=8	UCL	United Kingdom	high	2	2
10	National University of Singapore (NUS)	Singapore	high	19	13
11	The University of Hong Kong (HKU)	Hong Kong SAR	high	13	12
12	Nanyang Technological University, Singapore (NTU Singapore)	Singapore	high	24	20
13	Peking University	China (Mainland)	high	37	30
14	Tsinghua University	China (Mainland)	high	60	49
15	University of Pennsylvania	United States	high	11	7
=16	Cornell University	United States	high	5	4
=16	Yale University	United States	high	2	2
18	The Chinese University of Hong Kong (CUHK)	Hong Kong SAR	high	17	14
19	UNSW Sydney	Australia	high	3	2
=20	Johns Hopkins University	United States	high	8	6
=20	University of California, Berkeley (UCB)	United States	high	16	11
=22	EPFL	Switzerland	high	10	9
=22	The University of Melbourne	Australia	high	3	2
24	The University of Chicago	United States	high	2	1
25	Technical University of Munich	Germany	high	8	7
26	Fudan University	China (Mainland)	medium	24	18
27	Princeton University	United States	high	3	3
28	The University of Sydney	Australia	medium	2	2
29	Australian National University (ANU)	Australia	high	6	6
30	McGill University	Canada	high	0	0
31	Monash University	Australia	high	8	8
32	University of Toronto	Canada	high	12	10
33	The Hong Kong University of Science and Technology (HKUST)	Hong Kong SAR	medium	19	16
34	Université PSL	France	high	2	1
35	The University of Edinburgh	United Kingdom	high	5	3
36	Shanghai Jiao Tong University	China (Mainland)	high	43	36
37	King's College London	United Kingdom	high	0	0
38	Seoul National University	South Korea	medium	14	11
39	The University of Tokyo	Japan	medium	5	4

Rank	Institution	Country	Confidence	List A	List B
=40	The University of Manchester	United Kingdom	high	0	0
=40	The University of Queensland	Australia	high	3	3
42	Yonsei University	South Korea	medium	7	7
=43	Columbia University	United States	high	6	6
=43	Institut Polytechnique de Paris	France	high	2	2
=45	Northwestern University	United States	high	1	0
=45	The University of British Columbia	Canada	high	4	3
47	Zhejiang University	China (Mainland)	medium	43	21
48	Delft University of Technology	Netherlands	medium	1	1
49	University of California, Los Angeles (UCLA)	United States	medium	8	6
50	The Hong Kong Polytechnic University	Hong Kong SAR	medium	8	6
51	University of Michigan-Ann Arbor	United States	medium	6	5
=52	City University of Hong Kong (CityUHK)	Hong Kong SAR	high	10	7
=52	Korea University	South Korea	medium	3	3
54	National Taiwan University (NTU)	Taiwan	medium	4	4
55	Carnegie Mellon University	United States	high	19	17
56	Universiti Malaya (UM)	Malaysia	high	0	0
57	University of Bristol	United Kingdom	high	1	1
58	New York University (NYU)	United States	high	11	5
59	KU Leuven	Belgium	medium	1	1
60	University of Amsterdam	Netherlands	medium	6	6
61	Ludwig-Maximilians-Universität München	Germany	medium	3	3
62	The London School of Economics and Political Science (LSE)	United Kingdom	medium	0	0
63	King Fahd University of Petroleum & Minerals (KFUPM)	Saudi Arabia	high	0	0
64	Kyoto University	Japan	high	0	0
65	KAIST - Korea Advanced Institute of Science & Technology	South Korea	medium	21	18
66	Brown University	United States	medium	5	4
67	The University of Auckland	New Zealand	high	0	0
=68	University of Warwick	United Kingdom	high	0	0
=68	University of Birmingham	United Kingdom	high	1	1
70	Duke University	United States	medium	4	2
71	Lund University	Sweden	high	0	0
72	The University of Texas at Austin	United States	medium	13	10
73	Sorbonne University	France	medium	1	1
74	University of Illinois Urbana-Champaign	United States	medium	17	12
75	Trinity College Dublin, The University of Dublin	Ireland	high	0	0
76	Université Paris-Saclay	France	medium	0	0
=77	The University of Western Australia	Australia	medium	2	1
=77	University of Leeds	United Kingdom	high	0	0
79	Adelaide University	Australia	medium	4	4
80	University of Glasgow	United Kingdom	high	0	0
81	University of California, San Diego (UCSD)	United States	medium	10	8
=82	KTH Royal Institute of Technology	Sweden	high	1	1

Rank	Institution	Country	Confidence	List A	List B
=82	The University of Sheffield	United Kingdom	high	1	1
84	Universidad de Buenos Aires (UBA)	Argentina	high	0	0
85	Durham University	United Kingdom	high	0	0
86	Heidelberg University	Germany	medium	3	3
=87	Politecnico di Milano	Italy	high	0	0
=87	University of Technology Sydney	Australia	high	2	0
=87	Uppsala University	Sweden	high	0	0
=90	Nanjing University	China (Mainland)	medium	20	17
=90	University of Copenhagen	Denmark	high	2	2
=92	The Pennsylvania State University	United States	high	1	1
=92	University of Washington	United States	medium	11	8
94	Boston University	United States	high	5	5
95	The University of Osaka	Japan	medium	1	1
96	University of Alberta	Canada	high	1	1
=97	Institute of Science Tokyo	Japan	high	2	1
=97	University of Nottingham	United Kingdom	high	0	0
=98	Freie Universitaet Berlin	Germany	medium	0	0
=98	University of Zurich	Switzerland	high	0	0
=100	Purdue University	United States	high	8	4
=100	University College Dublin	Ireland	medium	0	0

Appendix B Data files, sources and licences

Files under data/ are in the private source repository; the website ships only the public data (docs/data).

File	Content	Source / licence
data/faculty/list_a_five_venues.csv, list_b_six_venues.csv	lists with venue presence, counts, topics, links	derived; see below
data/faculty/evidence_ledger.jsonl	atomic evidence records	derived
data/faculty/professor_dossiers.json	witness papers per venue, topic papers, positions	derived
data/rankings/qs_wur_2027_top100.json	QS 2027 ranks with sources and confidence	QS / university releases
data/papers/top_papers.json	240 curated accepted papers	proceedings, Paper Copilot, web
data/repos/github_repos.json	172 repositories	GitHub API
data/rules/writing_rulebook.json	12 venues, 148 rules with sources	venue sites and public writing advice
data/style/*	profiles, phrasebank, overlap index, manifest (no text)	derived from arXiv / Paper Copilot
Upstream: CSRankings	faculty affiliations and DBLP-derived counts	CC BY-NC-ND 4.0 (attribution: Emery Berger)
Upstream: DBLP	bibliographic records behind CSRankings	CC0
Upstream: Paper Copilot paperlists	accepted-paper metadata	no licence file; facts and links only